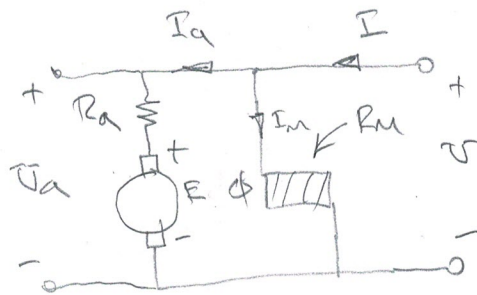


19.5)



$$R_a = 0.32 \Omega$$

$$R_m = 110 \Omega$$

NOM 457 A/R

$$I_{a0} = 6 A$$

$$n_0 = 1800 \text{ RPM}$$

$$V = 220 \text{ V DC}$$

BESTIM

a) WERT DER $I = 62 A$, DRAG 2V SPANNUNGSPFALL
OVER BORSSTAWA.

Lösung: WERT AN FÜR MOTOR GÄHRT AM

$$E = k_e \phi \omega$$

KSPL WER

$$-V + R_a I_a + E + V_B = 0$$

$$E = k_e \phi \omega = V - R_a I_a - V_B \quad (1)$$

$$k_e \phi = \frac{V - R_a I_a - V_B}{\omega}$$

$$= \frac{220 - 0.32 \cdot 62 - 2}{\frac{2\pi \cdot 1800}{60}}$$

$$= 1.1463 \quad (2)$$

WERT FÜR $I = 62 A$

I_m BILR

$$I_m = \frac{V}{R_m} = \frac{220}{110} = 2 A$$

$$I_a = I - I_m = 62 - 2 = 60 A$$

DRAG MED (2) AN (1)

$$\omega_1 = \frac{220 - 0.32 \cdot 60 - 2}{1.1463}$$

$$= 173.43 \text{ RAD/SEC} \approx \underline{1656 \text{ RPM}}$$

b) WERT

$$\underline{\text{SWAR:}} \quad \eta_1 = 1656 \text{ RPM/MIN}$$